

Kunj Sonker

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Education

Indian Institute of Information Technology Allahabad

B.Tech in Electronics and Communication Engineering

2024 – 2028(Expected)

CGPA: 7.85/10 (till 4th Sem)

Technical Skills

Languages:	Python, C++, C, Java, JavaScript (ES6+), TypeScript, HTML, CSS
Frameworks:	Node.js, Express.js, React.js, Next.js 15, Tailwind CSS, Framer Motion, Streamlit
Data Science:	scikit-learn, pandas, NumPy, DBSCAN, PyDeck, Geospatial Analytics, Haversine
Databases/Cloud:	MongoDB Atlas, PostgreSQL, MySQL, Firebase, Cloudinary, Streamlit Community Cloud
Tools:	Git, GitHub, Postman, JWT, Bcrypt, Multer, Axios, REST APIs, Vercel, Render
Soft Skills:	Analytical Thinking, Problem Solving, Self-Motivated, Detail-Oriented, Collaborative, Adaptable

Projects

ClearWay AI – Traffic Enforcement Intelligence System

[GitHub](#) | [Live](#)

- End-to-End ML Pipeline:** Built and deployed ClearWay AI solo — an end-to-end traffic enforcement intelligence system processing **300,000+ real parking violation records** spanning **12+ months** of historical data using **DBSCAN spatial clustering** with Haversine distance, detecting **18 active hotspots** from live Bengaluru data.
- Custom Scoring Model:** Designed a **Traffic Impact Score (TIS)** combining vehicle lane-blocking weight and obstruction duration to rank **18 enforcement zones** across **5+ violation categories**, shifting dispatch from random patrol to data-backed prioritization.
- Geospatial Dashboard:** Built and deployed a live **3D geospatial dashboard** using **Streamlit** and **PyDeck** with real-time filtering across **450+ km²** of Bengaluru's road network, hourly trend analytics, and one-click CSV export — publicly accessible on Streamlit Community Cloud.

Appointo – Full Stack Health-Tech Platform

[GitHub](#) | [Live](#)

- Multi-Role System:** Engineered **3** independent panels for Patients, Doctors, and Administrators, streamlining **100+** patient-doctor interactions with distinct dashboards, role-based access control, and dedicated workflows deployed on cloud infrastructure.
- Distributed Architecture:** Built a MERN backend using **MVC architecture** with **20+ RESTful API endpoints** to manage high-concurrency appointment scheduling across multiple user roles and services.
- Performance:** Achieved average API latency of **142ms** — **43% below** the industry-standard 250ms target — through optimized **MongoDB Atlas** indexing and JSON payload management (avg. 2.1KB).
- Security:** Implemented **stateless JWT authentication** and **Bcrypt** hashing with custom middleware enforcing role-based access control (RBAC) across all **3** panels.
- File Handling:** Integrated **Multer** for multipart file uploads and **Cloudinary** for scalable cloud-based storage and delivery of **100+** media assets.

Next.js Performance Portfolio

[GitHub](#) | [Live](#)

- Modern Web Stack:** Built using **Next.js 15 App Router** and **TypeScript**, achieving **96/100 Google Lighthouse** performance score and **100/100 SEO** score.
- Optimization:** Reduced **LCP to 0.8s** — **68% faster** than the 2.5s Core Web Vitals benchmark — via Server-Side Rendering (SSR), Next/Image optimization, and priority asset loading.
- GPU Rendering:** Engineered a custom particle interaction system using **HTML5 Canvas API** and **Framer Motion**, maintaining stable **60 FPS** through GPU-accelerated rendering.

Achievements

- Competitive Programming:** Solved **500+ problems** on LeetCode, CodeChef, and Codeforces; rated **Pupil** on Codeforces and **3-Star** on CodeChef; secured global ranks **96** (Div 3), **152** (Div 4), **237** (Div 2) among 65,000+ participants.
- Amazon HackOn'26:** Ranked in the **Top 75** coders nationwide in Amazon's premier coding competition.
- Flipkart GRiDlock' 26:** Qualified as a **Semifinalist** in Flipkart's flagship national engineering challenge.

Relevant Coursework

Data Structures & Algorithms (DSA)
Problem Solving (C/C++)

Microprocessor Interface
Linear Algebra

Network Theory